

Advance Analyzes for Blockchain Legal Issues in India

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Abstract:

This paper delves into the intricate legal landscape surrounding blockchain technology in the Indian context. It explores the potential challenges and opportunities that arise with the adoption of blockchain in various industries, shedding light on jurisdictional issues, smart contract enforceability, and the need for regulatory frameworks. The document emphasizes the importance of monitoring the evolving case law and legal principles related to blockchain technology in India, highlighting the necessity for a robust legal framework to address emerging issues. Additionally, it discusses the implications of blockchain technology on supply chain finance and the banking sector in India, offering insights into the barriers to adoption and potential solutions. Overall, the PDF serves as a valuable resource for legal scholars, policymakers, and industry professionals seeking to navigate the complex legal landscape of blockchain technology in India.

Keywords: Adoption barriers, Banking sector, Blockchain technology, Case law, Jurisdiction, Legal issues, Regulatory framework, Smart contracts, Supply chain finance.

1. Introduction

With the ever-changing landscape of technology, the Indian legal system has continually faced new challenges. This paper aims to explore the legal issues surrounding blockchain technology in India. Blockchain can be simply described as a decentralized digital ledger for the storage of data. A more technical definition is provided by the European Parliament in a 2017 publication which explains it as "a digital ledger in which transactions made in cryptocurrency are recorded chronologically and publicly". Blockchain technology was first devised in 2008 by "an unknown person or group of people under the pseudonym Satoshi Nakamoto who implemented it as a core component of the digital currency, Bitcoin" and it is best known as the technology underpinning Bitcoin. However, there are many other uses of this technology, including the possibility of vastly improving transparency, efficiency, and trust in multiple applications across the professional environment. Despite its many exciting opportunities and the serious challenges for the legal system, India's unique position has not yet been significantly explored in the analysis of blockchain technology. This paper will help to fill that gap. (Gad et al.2022) (Namasudra et al.2021) (Tanwar, 2022) (Gururaj et al.2020) (Kim, 2020)

1.1. Overview of blockchain technology

A blockchain is a decentralized digital ledger that records unalterable transactions. Each transaction generates a unique hash and is added to a block, creating a chain. Blockchain has transformative potential, extending beyond digital applications. It provides transparency and verification for various uses, like smart contracts and supply-chain transparency. Blockchain's key attributes are its decentralized nature and the use of hashes, making it virtually impossible to alter records. It can be used in industries such as finance, charities, and the public sector. However, data protection and potential consequences must be considered. The law surrounding blockchain is an area that will continue to expand alongside its use. (Schneider et al.2020) (Ralston, 2020)

1.2. Importance of analyzing legal issues in India

The study of blockchain is globally appealing and has gained attention from legal scholars and regulators worldwide. However, the legal issues surrounding blockchain in India have been overlooked. Despite not being widely embraced at the legislative level, there are certain restrictions on using blockchain in specific contexts, such as property or document records. Understanding blockchain technology beyond Bitcoin and virtual currencies reveals various legal issues, including data protection, smart contracts, representation, and jurisdictional problems. Organizations planning to utilize blockchain technology should consider the applicable laws and regulations, making it crucial to analyze legal issues in India. Blockchain's decentralized nature makes it difficult for a single entity to corrupt the ledger, potentially discouraging violators from non-compliance. Blockchain's core beauty lies in its digitized and decentralized public ledgers, which provide accurate and secure data. This research explores the legal aspects of blockchain, focusing on data protection, smart contracts, jurisdiction, and addressing challenges within traditional legal frameworks during the transitional period. It also examines the impact of blockchain on anti-money laundering and counter-terrorism financing regimes in Australia and highlights the need to address substantive legal issues in property regulations when employing blockchain technology. (Schuetz & Venkatesh, 2020) (Rana et al.2022) (Sharma and Joshi2021)

2. Regulatory Framework for Blockchain in India

The Indian felony device works in a manner that the right is taken into consideration as having been forbidden until it's far allowed through regulation. This subdivision analyses and examines the approaches the modern-day Indian felony system might be a barrier to the development of blockchain technology. It is also necessary to justify comments with the flow of the reasoning and substantiate feedback with relevant examples and/or authoritative blockchain literature. This may be done via the execution of advanced studies processes, along with assessing the criticisms and guidelines offered by blockchain specialists. Furthermore, comparative analyses among India and other international locations might be applied to show the effectiveness of shifting in the direction of a more

liberal legal regulatory machine for the improvement of blockchain technology in India. This might provide to demonstrate a right expertise to comment absolutely on a long way achieving criminal suggestions which might be required to help the operation and achievement of blockchain technology. Also, countries such as Estonia and Switzerland have adopted a more liberal mind-set in the direction of blockchain and that has led to a fulfillment of this technology within the provision of 'public services' along with the digital health record machine ('eleven', 2019). (Kaur et al.2022) (Patki & Sople, 2020) (Mathivathanan et al.2021) (Bhuvana and Aithal2020). Refer to Figure 1 for a detailed illustration of the blockchain structure, including its decentralized nature, hash mechanism, and transaction flow, along with insights into the current legal landscape for blockchain technology in India.

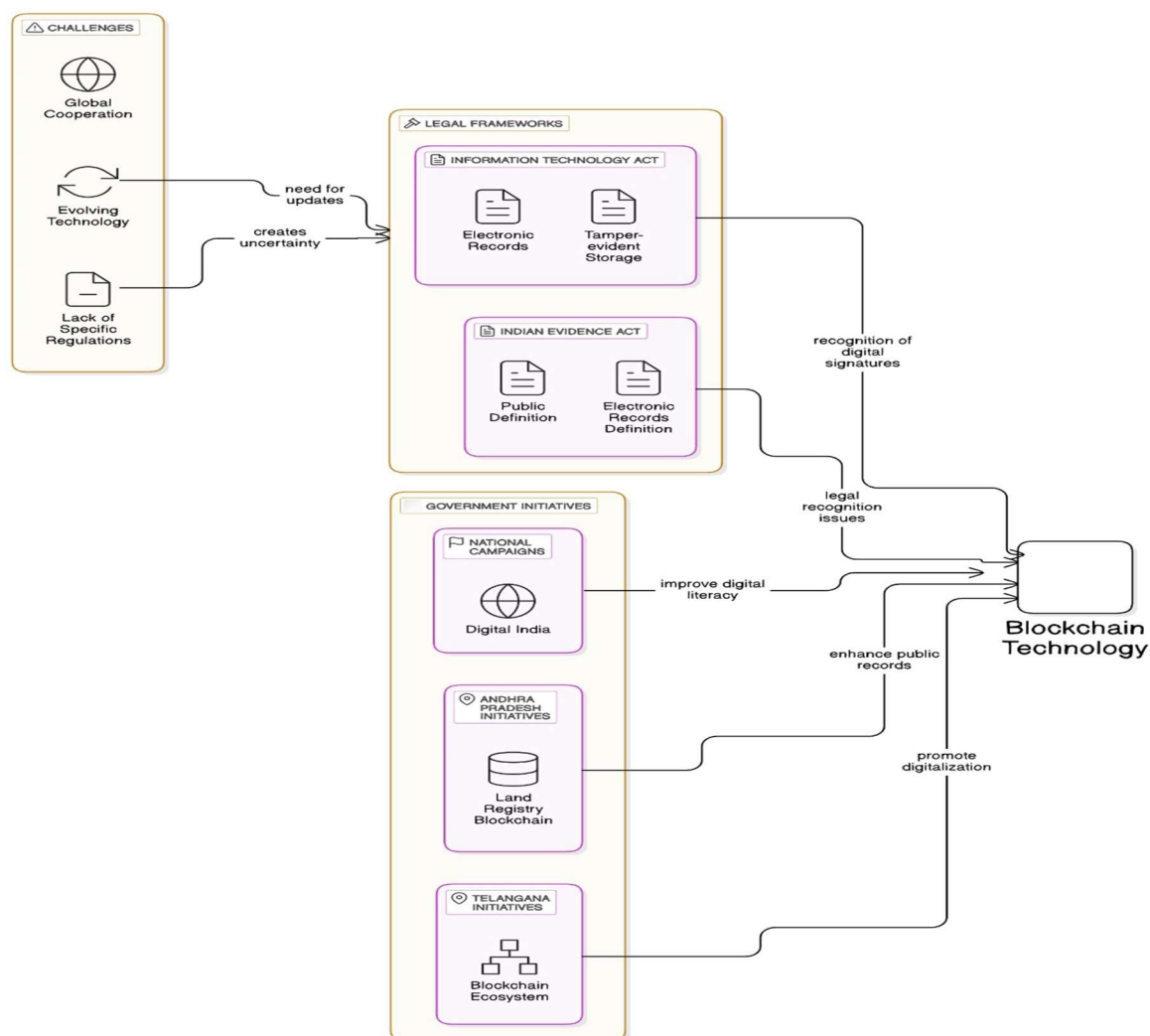


Figure 1: System Overview of Blockchain Technology in India

2.1. Current legal landscape for blockchain technology

Firstly, there is no express law or regulation on blockchain technology in India to date. As indicated above, most of the technology companies in India find their way to use blockchain technology under the Information Technology Act, 2000. Sections 91 and 92 of the Indian Evidence Act, 1872 dealing with the definition of the public at large and the definition of the electronic records and these definitions are the main issues when it comes to the legal recognition of blockchain technology in India. As per the Section 3 of the Information Technology Act, 2000, electronic records created through blockchain technology are recognized as the electronic record. But, there is no consensus or directly content which deals with the blockchain technology. It is not covered within the IT Act. According to the Rule 13 of the Information Technology (Preservation and Retention of Information by Intermediaries providing Digital Locker facilities) Rules, 2016, an electronic agreement is defined as an agreement which is concluded by affixing the digital signature in case of the parties have option of affixing digital signature. If such provisions are properly invoked for accepting the contents by affixing the digital signature or electronic signature then it is possible that a blockchain based agreement may become the electronic agreement. Also, the general practice in blockchain projects is that the provider in the chain transfer the data by way of the digital signature. So, if some regulations are prescribed for invoking these provisions then in the future the 'smart contract' can be recognized under the Indian laws. As per Section 13 of the Information Technology Act, 2000, when any information is being stored by the any person on any tamper-evident technology then such information shall be deemed to be a valid storage. It is a positive progress that the Government of Andhra Pradesh in India is trying to protect the tamper-evident technology based records through blockchain technology. The State of Andhra Pradesh has taken a lead where the state plans to protect the land records by creating a land registry on a platform of the blockchain. This is the first kind of initiative in India in which the government is trying to implement the blockchain technology for the public records. Also the states like Telangana and Maharashtra have shown their interests in the blockchain technology and they are creating a new block of records to start with. (Pandey & Sen, 2022) (Ghosh and Nanda2022) (Dimitropoulos, 2020) (Singh2020)

2.2. Government initiatives and policies

Different state governments in India recognize the potential of blockchain technology and its adoption on a global scale. They aim to promote the development of blockchain applications across industries to enhance the growth of the technology and related services in India.

On 26th September 2018, the Government of Telangana signed eight MoUs with consensus systems in New York. The objective is to create a blockchain ecosystem in Telangana and implement IT initiatives. The Consul General of India in New York played a crucial role in bringing the parties together. Dr. I Jayaraj, special chief secretary and IT advisor,

expressed excitement about the potential of blockchain technology for the state's digitalization.

The Andhra Pradesh government partnered with Covalent Fund to enhance cybersecurity, speed up services, and prevent cybercrime. Blockchain solutions will improve public services and governance reform.

The Indian government has been proactive in implementing blockchain technology in India. In 2016, the Indian government launched the "Digital India" campaign to ensure that government services are made available to citizens electronically. The main objective of the campaign is to connect rural areas with high-speed internet networks and improve digital literacy. Blockchain technology is recognized as an important research area by the Indian government and is funded by different government agencies such as the Department of Science and Technology. (Bhatt, 2020) (Gupta et al.2020) (Singh & Bansal, 2020) (Gulyani et al.) (Jain) (Beriya, 2021)

2.3. Challenges in implementing regulations

The challenges in implementing regulations are multi-pronged. Firstly, the technology is constantly evolving and it is difficult for the legislation to keep up. For example, courts need to understand complex technological concepts. Another complication is keeping regulations updated. The lack of industry-specific regulations in India also creates difficulty. Secondly, blockchain is part of a wider technological process, presenting complications for regulators. This may result in drafting broad legislations or implementing subsidiary legislations. This creates uncertainty for businesses. The paper will discuss learning from the European Union and the United States in protecting consumers' interests. Another problem is the decentralized nature of blockchain, making it difficult for countries to regulate data stored globally. Global cooperation is needed. This could be an obstacle to the effectiveness of the regulations. (Saraf et al.2024) (Kharitonova et al.2020)

3. Legal Issues and Challenges in Blockchain Adoption

The Algory project, initially registered with the Copyright Office, became the center of a legal dispute when its creator filed a lawsuit, alleging neglect of their research and testing contributions. The court ruled in favor of the creator, declaring that the defendants had no legal rights or interest in the invention (see Figure 2). Leaving research uncopyrightable would lead to less unlawful use of the technology. (Olson, 2020) (Kartiko and Soegiono2024) (Kingsley et al.2022)

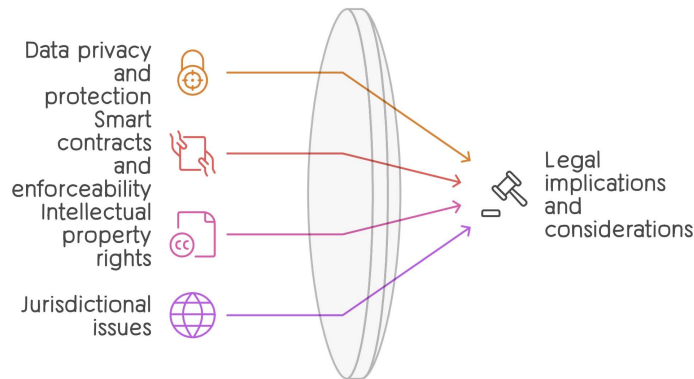


Figure 2: Legal Implications and Consideration

3.1. Data privacy and protection

It is no doubt to say that data security is one of the significant issues to the blockchain innovation. Distributing information between all members on a blockchain network makes it hard to revise or erase information if the need ought to emerge. It is likewise essential to take note of that the worldwide pattern of digitization has realized new ways information is being used and how that information is ensured. Truth be told, the requirement for information protection has become a worldwide worry with the execution of the General Data Protection Regulation in the European Union, which we will contrast with the present Indian lawful system for security of individual information. Hence, it is basic to comprehend the lawful commitments forced on any individual who is setting up a blockchain, yet additionally on any individual who is chosen as an information holder. Blockchain innovation ordinarily centers around the decentralized perspective and the straightforwardness of the information stream. Be that as it may, these features proposed by blockchain innovation may not generally hold up under a similarity to right security or consistence with the information security law. There are three key laws characterizing the space of information insurance and security in India: the IT Act 2000, the SPDI Rules 2011, and the novel Aadhar Act 2016. The centre rule of information assurance under the Indian enactment is to put the individual in charge of their own information and to guarantee the information is precisely put away and sensibly ensured, which is generally in accordance with the GDPR guideline in the EU. Assuming individual information neglects to be ensured, the law forces unforgiving punishments. (Guru et al., 2021) (Rao et al., 2021) (Simaiya et al.2020)

3.2. Smart contracts and enforceability

The concept of smart contracts is of utmost importance to blockchain technology as they represent the self-executing nature of a blockchain. Smart contracts are pre-programmable computer codes which are capable of executing on their own once certain predefined conditions are met. To put it simply, smart contracts are software that can automatically enforce and execute a contract when certain predefined conditions are met. However, as of

today, there is no specific legal framework available to recognize the legality and validity of such smart contracts in India. The Information Technology Act, 2000 which is the general statute governing cyber contracts in India, does not specifically deal with smart contracts. Moreover, the Indian Contract Act, 1872 as well does not specifically recognize or provide for the legality of smart contracts. It becomes important to ask the question - how the contracts executed by software will be recognized by the conventional legal framework when there is no specific statutory recognition to smart contracts. In the absence of any specific provision in the current law with respect to smart contracts, it becomes imperative to analyze how these self-executing digital contracts would be recognized and enforced under the current Indian legal framework and what are the potential legal challenges of the same. (Dwivedi et al.2021) (Drummer & Neumann, 2020) (Khan et al.2021)

3.3. Intellectual property rights

Intellectual property assets can be effectively managed using blockchain technology. Blockchains have the capability to transparently track and validate transactions and holdings of digital assets, which is particularly beneficial for intellectual property law. This includes managing financial interests, license rights, and providing a secure escrow facility for unregistered design or copyright material. However, implementing blockchain technology also requires establishing new rules and standards for its administration and control. This challenges traditional concepts and means of control in the intellectual property space. While there are potential benefits, such as redefining the legal 'author' of a work, there are also risks and costs involved, especially for error detection in digital assets. The use of blockchain to register patents or designs is no longer a technological pursuit but rather a consideration for legislation and practice to catch up with existing possibilities. Cybersecurity is another important aspect to consider, as personally identifiable information may be exposed in certain blockchain implementations. (Rambhia et al.2021) (Singh, 2022) (Bhadauria et al.2021) (Laroiya et al.2020) (& Krishna, 2022) (Chaudhary & Nidhi, 2023)

3.4. Jurisdictional issues

The development of case law and legal principles on blockchain technology in India will be an interesting area for legal scholars and the judiciary to watch in the coming years.

Similarly, as blockchain technology is adopted by more mainstream industries, it is conceivable that there will be legal issues when third-party rights are affected. For instance, where an individual or business seeks to rely on smart contract self-execution and a third party disputes the second element of a contract, namely that the contract is legally enforceable, there may be a need for courts to have jurisdiction. The issue of jurisdiction will become increasingly acute as blockchain technology becomes more prevalent in areas such as the internet of things and supply chain maintenance.

In international law, state sovereignty refers to the interrelated concept of territory and the right to rule. In the case of data created, stored, and transmitted on a blockchain, it exists on multiple computers in different jurisdictions simultaneously. It is unclear whether the Indian courts will have jurisdiction to settle a dispute. This leads to uncertainties for those in the blockchain industry and may undermine the attraction of India as a potential hub for blockchain innovation and business.

In India, there is as yet no legal definition of blockchain. The Indian legislature has avoided any direct reference to distributed ledger technology, the term "blockchain" or cryptocurrencies. However, section 2(1)(zb) of the Information Technology Act, 2000 ("IT Act") provides a definition of "electronic record" which states that data, record or data generated, image or sound stored, received or sent in an electronic form or microfilm or computer-generated microfiche. It also states that any reference to a law having been made in the electronic form will be construed as a reference to that law or regulation or rule and such statute, ordinance, etc. in force in India. This quite broad and generic definition could mean that the Indian courts will have the power to adjudicate in legal cases in respect of blockchain-related activities. However, as the technology and its application is a relatively new area, the question of the territorial jurisdiction of the Indian legal system becomes important.

Jurisdictional challenges in the field of blockchain technology relate to the nature of the technology itself. Blockchain is decentralized, which means that it operates on a global scale. In the absence of national regulations on blockchain, a key legal question is of identifying the proper jurisdiction over a dispute. It is a fundamental principle of any legal system to establish the powers of the authorities and courts of that jurisdiction. (Mohit, 2023) (Dylag & Smith) (SINGH, 2023) (Sharma et al.2021) (Dabas and Bhatt2022) (Jaya Prakash & Putchala, 2023).

4. Future Outlook and Recommendations

The Indian judiciary should promote the use of blockchain in the legal system. Allowing blockchain evidence in certain legal proceedings would establish India as progressive and technology-friendly. Creating a specialized regulator with experts from relevant fields would stimulate adoption in different sectors. India needs to streamline its legal framework and provide clarity and guidance to attract foreign investment in blockchain technology. The potential for blockchain in India is significant. (Haridas et al.2022) (Khairwal and Shah2020).



Figure 3: Future Outlook and Recommendation

4.1. Potential solutions for legal challenges

Firstly, it is vital to develop a comprehensive legal and regulatory framework to address the legal challenges discovered in the advancement of blockchain technology in India. This framework should be inclusive and ensure that every legal challenge is undertaken properly. The policy makers in India should adopt a new approach in policy design that will include all stakeholders and provide a test bed to trial and develop technology that can be regulated in the future. The government should continue with the momentum it has already started of adopting blockchain technology especially in land registration system, digital identity and electronic health records. This has two advantages; one it is expected that successful deployment of these technologies will provide a good reason to improve and advance current legal systems and secondly it will create a pipeline of technology lawyers, academics and policy makers that might think about innovative ways to regulate the technological advancements through research and development. There should be a focus on smart contracts in Indian legal system. It is submitted that a clarification under Indian law is needed to confirm that a smart contract has legal effect. One solution could be that the IT Act is amended to add provisions that explicitly recognize blockchain and permit electronic records to be maintained on a blockchain. In tandem, amendments could be added to recognize smart contracts under Indian law as proposed in the UK Jurisdiction Taskforce consultation. This would afford a set of rights to parties who enter into a smart contract and aid the enforceability of such contracts under Indian legal system. This is further supported by the fact that most of the land in India should be registered and backed by a guarantee as to the title which requires a robust and secure land registration system. This is a legal requirement and this should pave the way for the use of blockchain technology provided that the relevant enactments in India permit digital registration and the formation of smart contracts on a blockchain. Last but not least, India should also consider joining the New York Convention on the Recognition and Enforcement of Foreign Arbitral Awards. This will be a good step forward to deal with the legal challenge of jurisdiction when planning to implement blockchain technology in India. This is because

the ratification of this convention and the enacting of appropriate arbitration laws in India will mean that parties will be able to agree to resolve disputes using an international arbitration regime and it will allow foreign awards to be enforced in India. This will facilitate blockchain technology to be more easily adopted and accepted in India as it alleviates the fear of legal uncertainties during international business transactions and cross-border trade in the digital economy. (MGBEMENA2022) (Shalini et al., 2023)

4.2. Collaboration between stakeholders

Members of the blockchain community want a lobby group to advise the government on blockchain and cryptocurrency. They aim to engage with the government to negotiate regulations and establish legal clarity. Compliance with existing laws is required, but there is no specific regulation for blockchain and legal technology. Commentators suggest creating an industry advisory board to improve current laws. A regulatory sandbox for startups to work on new technologies, like blockchain and smart contracts, is also proposed. Responsive regulation that allows legislation to be influenced by technology use is mentioned in literature. A fuller understanding of a technology's efficacy will emerge through its development under the law's oversight. (Bhuvana and Aithal2020) (Rana et al.2022) (Thakur et al.2020)

Conclusion

India can take policy measures to promote and adopt blockchain technology. The first step is raising awareness about its utilities. The second step is hiring more consultants and providing training programs for advocates, judges, and officers. A regulatory sandbox can be created for experimenting with blockchain technology. A separate panel can be formed to manage and register blockchain development, with technical experts, government representatives, and academic consultants. This panel can grant registration for new blockchain solutions.ety or the trade. The panel ensures that proper independent work and innovation is appreciated in a secure and regulated manner.

References:

1. Gad, A. G., Mosa, D. T., Abualigah, L., & Abohany, A. A. (2022). Emerging trends in blockchain technology and applications: A review and outlook. *Journal of King Saud University-Computer and Information Sciences*, 34(9), 6719-6742.[sciencedirect.com](https://www.sciencedirect.com)
2. Namasudra, S., Deka, G. C., Johri, P., Hosseinpour, M., & Gandomi, A. H. (2021). The revolution of blockchain: State-of-the-art and research challenges. *Archives of Computational Methods in Engineering*, 28, 1497-1515.[researchgate.net](https://www.researchgate.net)
3. Tanwar, S. (2022). Blockchain Technology.[HTML](#)
4. Gururaj, H. L., Manoj Athreya, A., Kumar, A. A., Holla, A. M., Nagarajath, S. M., & Ravi Kumar, V. (2020). Blockchain: A new era of technology. *Cryptocurrencies and blockchain technology applications*, 1-24.[HTML](#)

5. Kim, J. W. (2020). Blockchain technology and its applications: Case studies. *Journal of System and Management Sciences*.aasmr.org
6. Schneider, S., Leyer, M., & Tate, M. (2020). The transformational impact of blockchain technology on business models and ecosystems: A symbiosis of human and technology agents. *IEEE Transactions on Engineering Management*, 67(4), 1184-1195.HTML
7. Ralston, S. J. (2020). Postdigital prospects for blockchain-disrupted higher education: beyond the theater, memes and marketing hype. *Postdigital Science and Education*.springer.com
8. Schuetz, S. & Venkatesh, V. (2020). Blockchain, adoption, and financial inclusion in India: Research opportunities. *International journal of information management*.vt.edu
9. Rana, N. P., Dwivedi, Y. K., & Hughes, D. L. (2022). Analysis of challenges for blockchain adoption within the Indian public sector: An interpretive structural modelling approach. *Information Technology & People*, 35(2), 548-576.brad.ac.uk
10. Sharma, M., & Joshi, S. (2021). Barriers to blockchain adoption in health-care industry: an Indian perspective. *Journal of Global Operations and Strategic Sourcing*, 14(1), 134-169.HTML
11. Kaur, J., Kumar, S., Narkhede, B. E., Dabić, M., Rathore, A. P. S., & Joshi, R. (2022). Barriers to blockchain adoption for supply chain finance: the case of Indian SMEs. *Electronic Commerce Research*, 1-38.springer.com
12. Patki, A. & Sople, V. (2020). Indian banking sector: Blockchain implementation, challenges and way forward. *Journal of Banking and Financial Technology*.HTML
13. Mathivathanan, D., Mathiyazhagan, K., Rana, N. P., Khorana, S., & Dwivedi, Y. K. (2021). Barriers to the adoption of blockchain technology in business supply chains: a total interpretive structural modelling (TISM) approach. *International Journal of Production Research*, 59(11), 3338-3359.bournemouth.ac.uk
14. Bhuvana, R., & Aithal, P. S. (2020). RBI distributed ledger technology and blockchain-a future of decentralized India. *International Journal of Management, Technology and Social Sciences (IJMTS)*, 5(1), 227-237.supublication.com
15. Pandey, S. & Sen, C. (2022). Blockchain Technology in Real-Time Governance: An Indian Scenario. *Indian Journal of Public Administration*.researchgate.net
16. Ghosh, J., & Nanda, A. S. (2022). Legal implication of blockchain technology in public health. In *Digital Health Transformation with Blockchain and Artificial Intelligence* (pp. 299-318). CRC Press.HTML
17. Dimitropoulos, G. (2020). The law of blockchain. *Wash. L. Rev.*.uw.edu
18. Singh, P. (2020, December). Role of blockchain technology in digitization of land records in Indian scenario. In *IOP Conference Series: Earth and Environmental Science* (Vol. 614, No. 1, p. 012055). IOP Publishing.iop.org
19. Bhatt, S. (2020). Digitalization of rural india: Digital village. *VISION: Journal of Indian Taxation*.researchgate.net

20. Gupta, R., Pal, S. K., & Muttou, S. K. (2020). Data analytics for better branding of e-governance and e-business systems: Case of "DIGITAL India" campaign. In *Leveraging digital innovation for governance, public administration, and citizen services: Emerging research and opportunities* (pp. 51-78). IGI Global.HTML
21. Singh, R. & Bansal, R. (2020). Benefits & Challenges of Digital India: A Study. Impact of COVID-19 on Global Economy.researchgate.net
22. Gulyani, M., Pasricha, C., Batra, P., Student, M. B. A., & Goel, C. (). A STUDY ON GROWTH OF DIGITAL INDIA CAMPAIGN. system.ijeast.com
23. Jain, M. D. (). A Study on Growth and Prospect of Digital India Campaign. ijaem.net.ijaem.net
24. Beriya, A. (2021). Digital India Programme: Going Full Circle.econstor.eu
25. Saraf, K., Bajar, K., Jain, A., & Barve, A. (2024). Assessment of barriers impeding the incorporation of blockchain technology in the service sector: a case of hotel and health care. *Journal of Modelling in Management*, 19(2), 407-440.HTML
26. Kharitonova, E., Kharitonova, N., Pulyaeva, V., & Shaidullina, V. (2020, October). Revealing best practices in legal regulation in the sphere of blockchain technology. In *Proceedings of the 2nd International Scientific Conference on Innovations in Digital Economy* (pp. 1-8).HTML
27. Olson, K. K. (2020). The Copyright Claims Board and the Individual Creator: Is Real Reform Possible?. *Communication Law and Policy*.HTML
28. Kartiko, N. D., & Soegiono, S. P. (2024). Analysis of the Copyright Dispute Decision Between Djanuar Ishak and PT. Elang Prima Retailindo: Study Of Commercial Court Decision Number 35/PDT. SUS-COPYRIGHT/2021/PN JKT. PUS. *Journal of Legal and Cultural Analytics*, 3(1), 105-116.formosapublisher.org
29. Kingsley, S., Sinha, P., Wang, C., Eslami, M., & Hong, J. I. (2022). " Give Everybody [...] a Little Bit More Equity": Content Creator Perspectives and Responses to the Algorithmic Demonetization of Content Associated with Disadvantaged Groups. *Proceedings of the ACM on Human-Computer Interaction*, 6(CSCW2), 1-37.acm.org
30. Guru, D., Perumal, S., & Varadarajan, V. (2021). Approaches towards blockchain innovation: A survey and future directions. *Electronics.mdpi.com*
31. Rao, U. P., Shukla, P. K., Trivedi, C., Gupta, S., & Shibeshi, Z. S. (2021). Blockchain for Information Security and Privacy.HTML
32. Simaiya, S., Lilhore, U. K., Sharma, S. K., Gupta, K., & Baggan, V. (2020). Blockchain: A new technology to enhance data security and privacy in Internet of things. *Journal of Computational and Theoretical Nanoscience*, 17(6), 2552-2556.researchgate.net
33. Dwivedi, V., Norta, A., Wulf, A., Leiding, B., Saxena, S., & Udokwu, C. (2021). A formal specification smart-contract language for legally binding decentralized autonomous organizations. *IEEE access*, 9, 76069-76082.ieee.org

34. Drummer, D. & Neumann, D. (2020). Is code law? Current legal and technical adoption issues and remedies for blockchain-enabled smart contracts. Journal of information technology.HTML
35. Khan, S. N., Loukil, F., Ghedira-Guegan, C., Benkhelifa, E., & Bani-Hani, A. (2021). Blockchain smart contracts: Applications, challenges, and future trends. Peer-to-peer Networking and Applications, 14, 2901-2925.springer.com
36. Rambhia, V., Mehta, V., Mehta, R., Shah, R., & Patel, D. (2021). Intellectual Property Rights Management Using Blockchain. In Information and Communication Technology for Competitive Strategies (ICTCS 2020) Intelligent Strategies for ICT (pp. 545-552). Springer Singapore.researchgate.net
37. Singh, S. (2022). IPR and Blockchain Technology. Issue 2 Int'l JL Mgmt. & Human..HTML
38. Bhadauria, S., Kumar, P., & Mohanty, T. (2021, December). Intellectual Property Protection using Blockchain and Digital Watermarking. In 2021 IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS) (pp. 1-6). IEEE.HTML
39. Laroia, C., Saxena, D., & Komalavalli, C. (2020). Applications of blockchain technology. In Handbook of research on blockchain technology (pp. 213-243). Academic press.researchgate.net
40. , K. .. & Krishna, V. a. n. a. t. h. i. (2022). Role of Intellectual Property in Blockchain. Part 2 Indian J. Integrated Rsch. L..HTML
41. Chaudhary, G. & Nidhi, A. (2023). Artificial intelligence and blockchain: a breakthrough collaboration in IP law. Journal of Intellectual Property Rights (JIPR).niscpr.res.in
42. Mohit, V. (2023). Regulating Block-Chain Technology in the Legal Field. Issue 2 Indian JL & Legal Rsch..HTML
43. Dylag, M. & Smith, H. (). From cryptocurrencies to cryptocourts: blockchain and the financialization of dispute resolution platforms. Information.tandfonline.com
44. SINGH, D. R. B. (2023). ... OF PRIVACY AND HEALTHCARE TREATMENT: PROTECTIVE MEASURES VIA APPLICATION AND EFFECTIVENESS OF BLOCKCHAIN TECHNOLOGY IN LEGAL A LANDMARK ON THE INDIAN CONSTITUTION.HTML
45. Sharma, S., Gamoura, S., Prasad, D., & Aneja, A. (2021). Emerging Legal Informatics towards Legal Innovation: Current status and future challenges and opportunities. Legal Information Management, 21(3-4), 218-235.HTML
46. Dabas, D., & Bhatt, H. (2022). Trajectory of Blockchain Technology in India: From Use Cases to Data Protection Regime. Sambodhi Indological Research Journal of LDII ISSN, 2249-6661.HTML
47. Jaya Prakash, B. & Putchala, N. R. (2023). The automation in the Indian judiciary. AIP Conference Proceedings.HTML

48. Haridas, S., Saroj, S., Maddala, S. T., & Kiruthika, M. (2022). LegalLedger–Blockchain in judicial system. In Innovative Data Communication Technologies and Application: Proceedings of ICIDCA 2021 (pp. 471-482). Singapore: Springer Nature Singapore.HTML
49. Khairwal, N., & Shah, R. (2020). Current Indian Judicial System: Issues and Blockchain Solutions. In IC-BCT 2019: Proceedings of the International Conference on Blockchain Technology (pp. 229-239). Springer Singapore.HTML
50. MGBEMENA, E. (2022). RECOGNITION AND ENFORCEMENT OF INTERNATIONAL COMMERCIAL ARBITRATION AWARD IN NIGERIA AND IN INDIA: A CASE STUDY. RENAISSANCE UNIVERSITY LAW JOURNAL (RNULJ), 2(1), 129-146.rnujournals.com.ng
51. Shalini, S., Sapre, A. A., & Shah, S. A. (2023). THE SHAKY APPROACH OF THE INDIAN JUDICIARY IN ENFORCING BILATERAL INVESTMENT TREATY AWARDS. Lex Humana (ISSN 2175-0947).ucp.br
52. Thakur, V., Doja, M. N., Dwivedi, Y. K., Ahmad, T., & Khadanga, G. (2020). Land records on blockchain for implementation of land titling in India. International Journal of Information Management, 52, 101940.HTML